Introduction

The present volume collects a set of contributions presented at the *Lisbon International Conference: Philosophy of Science in the 21st Century – Challenges and Tasks*, organized by the Centre for Philosophy of Sciences of the University of Lisbon (CFCUL), which took place between December 4th and 6th of 2013, in the Faculty of Sciences of the University of Lisbon. A selection of papers has already been published in two special issues of the international journal *Axiomathes* (2015, vol. 25 - Issues 1 and 2).

The conference intended to look at the major *challenges and tasks of philosophy of science in the 21st century*. This was the declared an aim of the conference which most of the papers addressed. However, even if, overall, the papers presented at the Lisbon Conference covered a large spectrum of general, classical topics and approaches of the contemporary philosophy of science, such as logic, epistemology and argumentation, as well as different particular cases of disciplinary approaches, the fact is that a special emphasis was given to some specific questions. Among these were the relationship between metaphysics and science, the requirement of a true critical scientific realism, the need of a statistic control of big data, the requirement of overcoming emotions in argumentation, the aim of rethinking the role of fiction in explanatory devices, the critical concern about the effects of the new media communication on scientific activity or the apprehension regarding the future of science and civilization in the dark days we are living in. As the reader will notice many of these questions appear in the articles collected in the present book.

The volume opens with three papers that deal with some of the main issues concerning the internal relations between philosophy of science and metaphysics.

In a much inspiring and ambitious paper, Jan Sebastik addresses some of the most important "Turning Points in the Development of Science", particularly of mathematics, astronomy and physics, in relation to philosophy and to the general picture of the world. After presenting the historical foundations of European civilization on both Greek and Islamic scientific cultures, and examining the reasons for their decline, Sebastik addresses the birth of modern science in the 17th century, focusing on the work of Kepler and on his collaboration with Tycho Brahe. As the author observes, "[w]ithout the couple Brahe-Kepler, or with new observational data, we perhaps never would have Newton's laws and would have just local regularities or a collection of isolated laws like in China or India, without a picture of the universe unified by universal laws." Nevertheless, despite the acknowledgement that "no great civilization except Europe developed objective rational and systematic knowledge that combines mathematics with experiments", Sebastik warns that today we need to critically reflect upon the first warning signals of recession which are manifest in European and American's scientific culture and education. Indeed, "Science is not only our work and our heritage; it is also the only means to keep pace with quickly developing outer world."

Acknowledging a growing resurgence of metaphysics in the philosophy of science in the last decades (after logical empiricism and the Vienna Circle), Nuno Fonseca focuses on the recent debate over the possibility of a scientific metaphysics and the need for a metaphysics of science. In his "Metaphysics and Science: With or Without? Clearing up for an inclusive disjunction", Fonseca argues that although we should be aware of the distinctiveness of both domains and of the dangers of an excessive mutual acquaintance, one of the challenges for the 21st century Philosophy of Science is to find a balanced complementarity between metaphysics and science, from which both would have much to gain.

In his "Carnap's Reconstruction of Theoretical Content and Structural Realism. A Reply to Friedman", Angelo Cei agrees with Friedman's view that Carnap's use of Ramseification represents part of his project of the Logic of Science. According to Cei, the contemporary debate on the viability of Structural Realism and the appropriate notion and characterization of structure explains the actuality and pertinence of rethinking Carnap's approach. Nevertheless, Cei questions Friedman's interpretation, according to which Carnap's endeavour succeeds in achieving an ontologically neutral form of Structural Realism. The author then proposes what he considers to be a desirable feature of any neutralism compatible with Structural Realism's main goal, showing that such a feature cannot be obtained by Carnap's account of the distinction between analytic and synthetic truths. As Cei argues, Carnap's structuralism does not allow us to see in what sense there is continuity through theoretical change.

The second part of this volume collects four articles addressing distinct logical and epistemological issues in the contemporary philosophy of science.

Assuming that there may be a common underlying logic that regulates the inferential activities of every scientific practice, Angel Nepomuceno-Fernández attempts to use logical dynamics of information and communication (as presented by van Benthem) by means of the family of systems of dynamics epistemic logic, and applying it to explain the non classical features of quantum systems. Moreover, according to Nepomuceno-Fernández's "Quantum Logic from a Dynamic Perspective", traditional quantum logic can be subsumed in a propositional dynamic quantum logic, and dynamic logic may be regarded as a useful underlying logic of physical practices.

In "The Ways of Probable Truth", Dinis Pestana and Fernando Sequeira observe that since statistic is a necessary tool for scientific research, we need to determine the ways to establish the probable truths. Otherwise, claim the authors, we are condemned to make bad science. Hence, Pestana and Sequeira present and compare some examples of good and bad science built upon an intensive use of statistics. This leads them to emphasise that the power of statistics stems from its usefulness in rejecting false conjectures, with a caveat: few data can lead to a biased view, but too much data may exhibit irrelevant statistical results.

In "The Headscarf Debate: Surprise and Meta-emotions in Argumentation", Dina Mendonça aims to show the importance of taking into account the complexity of emotional processes in order to achieve a deeper understanding of the role of emotions in reasoning and argumentation processes. According to the author, the analysis of the so-called headscarf debate (particularly after the French law that forbs students to wear visible religious signs in public schools, in 2004, and the law that bans the burqa and the niqab as garments to be used in public, in 2010) can show the way in which the emotion of surprise uncovers different emotional layers in argumentation such as meta-emotions, masked emotions and existential emotions. As Mendonça states, argumentation theory should be regularly updated with the developments of research in emotion theory.

The last essay in this second part confronts us with a challenging question: "Are Scientific Thought Experiments Objects of Fiction?". According to Fernando Rua, thought experiments constitute a challenge to the received theories of explanation, since they have completely distinct starting points. Whereas thought experiments try to achieve new epistemic conclusions based on imaginary constructions, using just past knowledge, theories of explanation are usually based on the representational method of collecting information. The author then presents the guidelines of a new fictionalist kind of approach able to explain the nature of thought experiments, which involves a commitment to a constructivist theory of knowledge and a re-examination of notions such as existence, possibility, imaginary entities and (Meinongian) homeless objects.

The third part of the book presents five contributions to different areas of disciplinary research.

In his "Imagination and visualization of geometrical and topological forms in space. On some formal, philosophical and pictorial aspects of mathematics", Luciano Boi assumes two main goals: to reveal the theoretical role of intuition in the practice of mathematics discovering, and to stress the key role visualization plays in all the reasoning steps about mathematical objects and in the demonstration of their properties. Furthermore, according to Boi, visualization can show us the "concrete" existence of a new class of mathematical objects, such as Riemann surfaces and minimal surfaces. Therefore, says the author, "if we want shed new light on some crucial mathematical issues and reach a deeper understanding of the relationship between mathematics and the real world, as studied by natural as well human science and art", the visualization method in mathematics should be discussed and further developed.

In "The Crisis in Economic Theory. Shifting to a new paradigm in Economics?", Ângela Dionísio notes that the economists' failure to predict and find a solution to the financial crisis of 2007-2008 has exposed the limits and ultimately the inadequacy of the mathematical, axiomatic and deductive abstract formal models employed by mainstream or neoclassical economy. According to the author, although a new economic thinking is now emerging (for example, a new complex systems' view confronts the old reductionist approaches), the question is whether we can affirm that it corresponds to a genuine shift in the economic paradigm (in Kuhnian terms), or if neoclassic economic paradigm is being replaced as the dominant research program (in Lakatian terms) in economy.

In "Philosophie de l'Internet", Éric Guichard develops an analysis of the global system of interconnected computer networks, commonly called Internet, in the light of a philosophy of technology. He takes into consideration its social, cultural and epistemological effects, mainly at the level of what Guichard calls the 'culture of the writing activity' that is proper to Internet ("*la culture de l'écrit propre à l'internet*")'. From a three-dimensional epistemological perspective, based on the connections between experience, theory and technique, Guichard complements the first part of his philosophical and anthropological study, with the development of a more externalist approach, reflecting upon the effects of this "new industry of the writing process" at the Universities.

Filipe Varela's "Body, Topological Deformation, Desire" addresses an ensemble of questions that concern our relationship with the human body, in a time that it is permanently subject to processes of reshaping it by modern technology (breast implants, liposculpture suction lipectomy, videogames software's reshaping our body's topology, etc.). Thus, asks Varela: What conceptions of desire does this re-writing of the body summon? Is there any desire at all, or are we facing sheer curiosity? Alternatively, are we embracing a teleology of the excess through shape's transformation that Bataille's eccentric morphology suggested? Another group of questions concerns the problems of identity and intelligibility. Can mathematics of morphology or a topology (not a geometry) of desire bring intelligibility to this exploration of the human body? Still, according to the author the main point is an ontological one: does shape's deformation affect the form of being?

The last article of this section, "The Good and the Goods in Philosophical Tradition and Contemporary Ethics", from Paulo de Sousa Mendes, addresses the notion of good and the different ways it was conceptualised in the history of thought. According to the author, modern age, discussing the idea of a supreme good in human life beyond the context of religion and revelation, ended up by narrowing the question of the good, reducing it to external goods and to the erroneous idea that the liberal and democratic society should only provide the protection of such kinds of goods. In Ethics, the split within modernity occurred between liberty (e.g., Kant) and utility (e.g., Bentham). Sousa Mendes concludes his brief essay advocating the necessity of returning to a broad conception of the good and the goods similar to the Aristotelian tradition.

The last two essays envisage, in a different way, the past and the future of the philosophical and scientific rationality. In "Mito, Ciência, Literatura: A Deriva dos

Continentes", António Bracinha Vieira, a theoretical biologist whose work has mainly concentrated in biological anthropology and palaeoanthropology, develops a personal reflection about the common roots and the differences between natural sciences, literature, philosophy, logic, mathematics, law, religion and myth, and art.

The last article of this book, written by Zbigniew Kotowicz, evokes a philosopher that has been present from the very beginning of CFCUL's activities: Gaston Bachelard. In "Gaston Bachelard: Epistemology, History", Kotowicz stresses that more than a philosopher of science, *stricto sensu*, Bachelard is rather an explorer of the 'workings of scientific rationality', whose writings evolve around two axes. The first corresponds to a scrutiny of the various forms of the past scientific rationality. The second manifests itself in the form of a reflection about the challenges set by the twentieth century science for philosophical thought. After analysing the question of the relation between epistemology and history of science in Bachelard's work, the author concludes: "Bachelard's reflections suggest that there is only an apparent overlap between the philosophy of science and history of science. In fact, they may be in a sense opposite. The historian looks to the past: the epistemologist is on the lookout for the new".

The 2013 *Lisbon International Conference: Philosophy of Science in the 21st Century – Challenges and Tasks* was an exciting and emotive event. CFCUL was at that time suffering the effects of a crude financial crisis, which was undermining Portuguese society, Portuguese university and Portuguese research. CFCUL had limited resources for putting forward the Conference. However, it did not prevent the presence of more than one hundred participants from all over the world, both students, researchers and key note speakers who accepted to came by their own expenses, and without any material support provided by the organizers of the conference. It was beautiful to see the commitment of all in the celebration of our common interest in philosophy of science. To all we would like to express our gratefulness.